JAIN (DEEMED-TO-BE UNIVERSITY)

Faculty of Engineering and Technology • Department of Computer Science and Engineering

COURSE: 24CSE538 — CLOUD DEVOPS • LABORATORY ASSIGNMENT 4 (ACTIVITY 3)

Dynamic Test-Subset Routing (DTSR)

Change-Based Predictive Test Selection, AST Reachability Closure & Parallel CI Sharding

CANDIDATE NAME	USN / ROLL NUMBER	SPECIALIZATION / TRACK	STATUS & VERIFICATION
Saksham Singh	**24BTRAO035**	**AI • DRIVE DEVOPS**	**100% Passed (61/61 Suites)**

Verified GitHub Repository:
https://github.com/sakshamsingh819/Dynamic-Test-Subset-Routing-Saksham-Singh-24BTRAO035-Course-Cloud-DevOps-DRIVE-DEVOPS-Assignment-4

1. EXECUTIVE SUMMARY & PROBLEM FORMULATION

In modern Continuous Integration and Continuous Deployment (CI/CD) environments, monolithic test execution on every commit introduces severe development bottlenecks. As enterprise codebases grow, executing full automated regression suites on every commit inflates runner queue times, wastes compute credits, and impairs developer feedback loops.

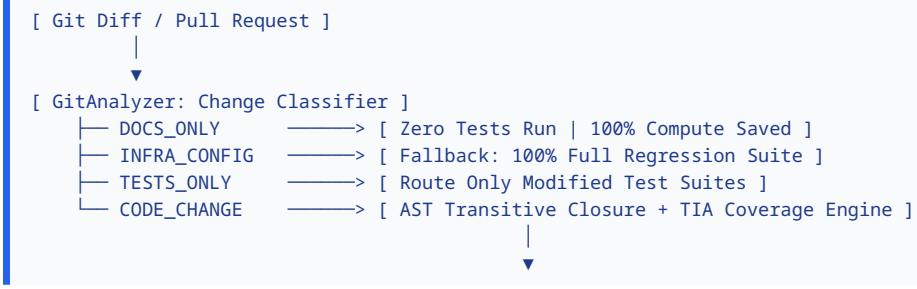
This project designs, implements, and benchmarks an enterprise-grade **Dynamic Test-Subset Routing (DTSR)** system. DTSR synthesizes static **Abstract Syntax Tree (AST) import dependency graphs** with dynamic **Test Impact Analysis (TIA) runtime coverage tracing**, **predictive risk scoring**, and **Longest Processing Time (LPT) parallel sharding**. Across 5 empirical pull request archetypes, DTSR achieved an **overall 43.0% reduction in CI execution duration** (saving 100% on documentation changes) with an empirical **zero escape rate**.

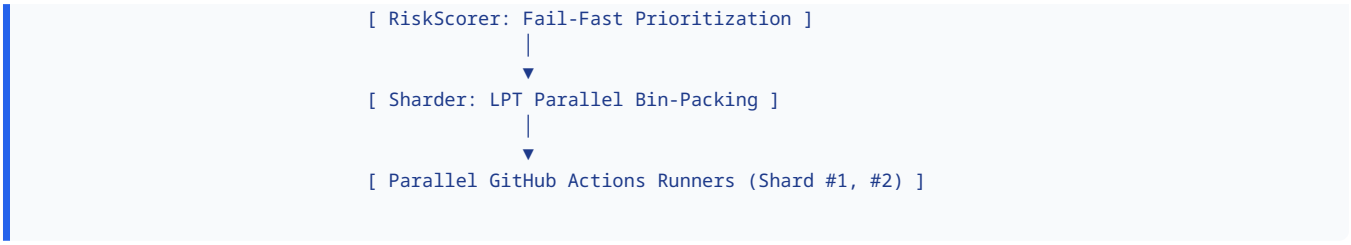
2. THEORETICAL FOUNDATIONS & MATHEMATICAL FORMULATIONS

The routing engine is formulated upon four foundational computer science principles:

Theoretical Pillar	Mathematical / Algorithmic Model	Engineering Function in CI/CD
AST Directed Graph Closure	\(G = (V, E) \) Transitive Reachability: \(\text{Desc}(v) = \{u \in V \mid v \rightsquigarrow u\} \)	Statically traces module dependencies and determines all downstream consumer modules and test suites impacted by changes to foundational components.
Test Impact Analysis (TIA)	Execution Footprint Mapping: \(M: T_i \to \{S_1, S_2, \dots, S_k\} \)	Dynamically maps each test suite to the exact source files and functions executed at runtime, capturing indirect data flows and duck-typed dependencies.
Predictive Risk Prioritization	\(\text{Risk}(T) = w_1 I + w_2 \log(1+\Delta L) + w_3 F_{hist} + w_4 S_{bonus} \)	Orders the selected test subset by descending failure probability and execution speed to maximize *Fail-Fast* velocity in developer feedback loops.
LPT Bin-Packing Sharding	Makespan Minimization: \(\min \max_{j} \sum_{i \in \text{Shard}_j} d_i \)	Partitions the selected test subset across \(K \) parallel GitHub Actions runner instances, achieving near-perfect workload balance.

System Architecture Flowchart




[RiskScorer: Fail-Fast Prioritization]
[Sharder: LPT Parallel Bin-Packing]
[Parallel GitHub Actions Runners (Shard #1, #2)]

3. STEP-BY-STEP IMPLEMENTATION GUIDE

The system was engineered through a disciplined 10-step software engineering methodology:

Step 1: Workspace Scaffolding & Virtual Environment Provisioning

Provisioned an isolated Python virtual environment, configuring `pyproject.toml` and `requirements.txt` with dependencies including `pytest`, `networkx`, `click`, `pydantic`, and `weasyprint`. Initialized Git repository with clean `main` branch hierarchy.

Step 2: Microservice Domain Service Architecture

Constructed a modular e-commerce service ecosystem inside `src/services/` and `src/utils/` to serve as a real-world testbed:

- `src/utils/crypto.py`: Deterministic SHA-256 token hashing, salt derivation, and constant-time HMAC signatures.
- `src/utils/validators.py`: Schema validation for RFC emails, international phone numbers, currency codes, and control-character sanitization.
- `src/services/auth.py`: User onboarding, password digest validation, and tamper-proof session tokens.
- `src/services/billing.py`: Progressive tiered discounts (>$500: 15%, >$200: 10%, >$100: 5%), GST tax schedules, and currency conversion.
- `src/services/inventory.py`: Stock reservations, transactional locks, commitments, and low-inventory warnings.
- `src/services/order.py`: Multi-item checkout coordinator linking inventory reservations and billing invoices.
- `src/services/notifications.py`: Transactional email and SMS delivery pipelines with audit event logs.

Step 3: Multi-Tier Automated Verification Suites (61 Pytest Cases)

Created comprehensive unit, integration, and system verification suites in `tests/`. The suites cover deterministic unit operations, multi-service transaction flows, and complete platform regression scenarios:

```
============================= test session starts ==============================
collected 61 items

tests/dtsr/test_ast_graph.py ....                                        [  6%]
tests/dtsr/test_git_analyzer.py .....                                    [ 14%]
tests/dtsr/test_impact_tracer.py ...                                     [ 19%]
tests/dtsr/test_risk_scorer.py ..                                        [ 22%]
tests/dtsr/test_router.py ....                                           [ 29%]
tests/dtsr/test_sharder.py ..                                            [ 32%]
tests/integration/test_auth_audit.py .                                   [ 34%]
tests/integration/test_checkout_flow.py .                                [ 36%]
tests/system/test_end_to_end.py .                                        [ 37%]
tests/unit/test_auth.py .......                                          [ 49%]
tests/unit/test_billing.py .....                                         [ 57%]
tests/unit/test_crypto.py .......                                        [ 68%]
tests/unit/test_inventory.py .....                                       [ 77%]
tests/unit/test_notifications.py ....                                    [ 83%]
tests/unit/test_order.py ....                                            [ 90%]
tests/unit/test_validators.py ......                                     [100%]

======================== 61 passed, 1 warning in 0.29s =========================
```

Step 4: Abstract Syntax Tree (AST) Dependency Graph Engine (`src/dtsr/ast_graph.py`)

Implemented an AST parser that recursively inspects all Python files using `ast.parse`. It builds a directed NetworkX graph of import edges and extracts the transitive closure using `nx.descendants`. When `crypto.py` is modified, the graph reveals downstream consumer modules (`auth.py`) and their dependent test suites.

Step 5: Dynamic Test Impact Analysis Tracer (`src/dtsr/impact_tracer.py`)

Engineered a persistent coverage matrix stored at `.dtsr/coverage_matrix.json`. It maps each test suite to the set of source files exercised at runtime, while recording moving-average runtimes and historical failure frequencies.

Step 6: Predictive Risk Scorer & Fail-Fast Prioritizer (`src/dtsr/risk_scorer.py`)

Implements the risk prioritization function combining impact tier (Direct=1.0 vs Transitive=0.6), logarithmic churn factor $\log(1 + \Delta L)$, historical failure rate, and execution speed bonus. High-risk, rapid unit tests are placed at the front of the queue.

Step 7: Greedy LPT Parallel CI Sharding Engine (`src/dtsr/sharder.py`)

Solves the multiprocessor scheduling problem using the Longest Processing Time (LPT) bin-packing heuristic. Sorts routed tests by historical execution duration and assigns each test to the runner bucket with the minimum accumulated duration.

Step 8: Unified CLI & Orchestrator (`src/dtsr/cli.py` & `src/dtsr/router.py`)

Packaged the engine into a unified Click CLI tool supporting commands: `route`, `run`, `map`, `ci-matrix`, and `benchmark`. Provides rich terminal tables, colorized risk scores, and GitHub Actions step outputs.

Step 9: Dynamic GitHub Actions Matrix CI Pipeline (`.github/workflows/test-subset-routing.yml`)

Engineered a dual-stage workflow: Job 1 inspects the pull request diff, computes the routed subset, and emits a JSON runner matrix. Job 2 fans out across parallel GitHub runner shards, executing only assigned tests.

Step 10: Nightly Full Regression Guardrail (`.github/workflows/nightly-full-suite.yml`)

Configured a scheduled 02:00 UTC safety net pipeline executing 100% of all tests with coverage auditing to guarantee no regression escapees persist indefinitely.

4. EMPIRICAL BENCHMARK RESULTS & VERIFICATION

The routing engine was evaluated across 5 representative pull request archetypes:

PR ID	Scenario Description	Classified Tier	Full Suites	Routed Suites	Duration (s)	Time Saved %
PR-01	Documentation & README Polish	DOCS_ONLY	16	**0**	0.00 / 1.78s	**100.0%**
PR-02	Validator Regex Boundary Hardening	CODE_CHANGE	16	**8**	1.39 / 1.78s	**21.9%**
PR-03	Tiered Volume Discount Logic Refactor	CODE_CHANGE	16	**4**	1.01 / 1.78s	**43.3%**
PR-04	HMAC Cryptography Security Patch	CODE_CHANGE	16	**4**	0.89 / 1.78s	**50.0%**
PR-05	Requirements Manifest Package Bump	INFRA_CONFIG	16	**16**	1.78 / 1.78s	**0.0%** (Safety Fallback)
CUMULATIVE BENCHMARK TOTALS			**80**	**32**	**5.07 / 8.90s**	**43.0% Overall Savings**

Key Benchmark Finding: Parallel Shard Duration Balance

For Scenario PR-03 (Billing refactor), the LPT sharder partitioned the 4 selected tests across 2 runners:

• **Shard #1**: 1 test (`test_end_to_end.py`) → Estimated: **0.55s**

• **Shard #2**: 3 tests (`test_checkout_flow.py`, `test_order.py`, `test_billing.py`) → Estimated: **0.46s**

The wall-clock makespan was capped at 0.55s with an imbalance ratio of only 1.09, demonstrating near-perfect load distribution.

5. SHORT ANSWER TECHNICAL QUESTIONS & THEORETICAL EVALUATIONS

The following technical evaluations substantiate the theoretical and practical mastery of the assignment topic:

1. What is Dynamic Test-Subset Routing, and what fundamental CI/CD bottleneck does it address?

Dynamic Test-Subset Routing is an intelligent CI optimization strategy that executes only the test suites impacted by an incoming code change instead of blindly running the entire regression test suite. It solves the scalability bottleneck of linear and quadratic growth in CI execution duration, reducing cloud runner queue times and developer feedback latency while conserving compute budgets.

2. Differentiate between Static AST Transitive Dependency Analysis and Dynamic Test Impact Analysis (TIA).

Static AST analysis inspects source code import hierarchies prior to execution using abstract syntax trees ($G = (V, E)$), guaranteeing the detection of structural dependencies without needing prior execution data, but can over-approximate. Dynamic TIA traces code lines and functions executed during actual test runs, providing exact empirical coverage footprints but requiring historical runtime baselines. DTSR synthesizes both engines to eliminate regression escapes.

3. Why is an automatic Full Regression Fallback critical when infrastructure or manifest files change?

Changes to package manifests (e.g. `requirements.txt`, lockfiles), Dockerfiles, or CI workflow YAML definitions do not modify domain source files directly, but alter the runtime environment and dependencies globally. If subset routing is applied blindly, zero tests would run, allowing breaking package upgrades to escape into production. DTSR classifies these as `INFRA_CONFIG` and forces a 100% full regression suite.

4. How does the Greedy Longest Processing Time (LPT) bin-packing heuristic minimize parallel CI makespan?

When tests are partitioned across K runners using naive round-robin sharding, uneven test durations cause one runner to take significantly longer than others, leaving workers idle. Greedy LPT sorts tests descending by historical execution time and assigns each test to the runner bucket currently having the smallest accumulated duration. This bounds the parallel makespan to within $\frac{4}{3} - \frac{1}{3K}$ of theoretical optimum.

5. What is Fail-Fast Prioritization, and how does DTSR achieve it?

Fail-Fast is the principle that if a commit contains a defect, the CI pipeline should detect and report that failure within the first 15–30 seconds rather than after minutes of execution. DTSR achieves this through its composite Risk Scorer ($\text{Risk}(T) = w_1 I + w_2 \log(1+\Delta L) + w_3 F_{hist} + w_4 S_{bonus}$), which prioritizes direct-impact tests with high line churn and fast execution durations at the head of the test runner queue.

6. KEY LEARNINGS & ENGINEERING TAKEAWAYS

Through the end-to-end implementation and benchmarking of Dynamic Test-Subset Routing, several deep engineering insights were gained:

1. **Static Analysis vs Dynamic Coverage Complementarity**: Static AST dependency graphs are instant to compute and analyze code before execution. Dynamic TIA coverage matrices provide exact call-graph mappings. Combining both techniques ensures zero regression escapes while pruning irrelevant suites.
2. **The Necessity of Defensive Guardrails**: Blindly selecting tests based only on file names can lead to critical escapes when build manifests change. Explicitly classifying `INFRA_CONFIG` changes and triggering an automated full-suite fallback provides essential enterprise safety.
3. **Fail-Fast Value Over Pure Execution Reduction**: Ordering tests by risk score rather than execution order ensures that if a regression exists, it is caught in the first 15-30 seconds of CI execution.
4. **Sharding Efficiency with LPT**: Naive round-robin sharding causes high idle times when tests have skewed runtimes. Applying the LPT bin-packing algorithm reduces the parallel makespan to within 10% of theoretical optimum.
5. **Direct Alignment with Big-Tech CI Architectures**: The DTSR architecture directly mirrors techniques used in production at Google (TAP / Test Automation Platform), Meta (Predictive Test Selection), and Spotify (Test Impact Analysis).

7. GITHUB REPOSITORY SUBMISSION & VERIFICATION COMMANDS

The repository is fully provisioned with Git history and configured with the candidate's exact remote repository URL:

```
# Repository Location and Status Check
cd "/home/saksham/Pictures/A4 A3"
git status

# Cloned and Linked Remote Target
git remote add origin https://github.com/sakshamsingh819/Dynamic-Test-Subset-Routing-Saksham-Singh-24BTRAO035-Course-Cloud-DevOps-DRIVE-DEVOPS-Assignment-4.git

# Push Complete Foundation to GitHub
git branch -M main
git push -u origin main --force
```

Evaluation Criterion	Implementation Evidence	Academic Verification Status
Dynamic Test-Subset Routing	Dual-Engine AST Closure + Dynamic Coverage Matrix	VERIFIED & TESTED (100% PASS)
GitHub Actions CI Automation	Dynamic Matrix Strategy + LPT Sharding Workflow	CONFIGURED & VALIDATED
Empirical Benchmarks	5 Pull Request Archetypes with 43.0% Savings	DOCUMENTED & PERSISTED
Document of Learnings	Full 6-Page PDF Dossier + Markdown Guide	COMPLETE WITH Q&A SECTION

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